

Two cases of QT prolongation with Cardizen in older adults

Abstract. We report two cases of marked QT-interval prolongation temporally associated with Cardizen, a fictional antiarrhythmic, in patients over 70.

Introduction. Methods. This was a single-centre observational report. Data were abstracted from the electronic health record by two independent reviewers and cross-checked, with discrepancies resolved by consensus. Adverse event terms were coded to MedDRA preferred terms (version 27.0). Causality was assessed with the WHO-UMC system and, independently, with the Naranjo probability scale. Written informed consent was obtained from the patient for publication of anonymised clinical details and images. Laboratory reference ranges are those of the reporting institution's accredited laboratory. Concomitant medicines, over-the-counter products and herbal preparations were recorded where documented. Baseline organ function was reviewed and prior exposure to the suspect class was sought from the dispensing history. The report follows the CARE guideline for case reports. No external funding was received and the authors declare no competing interests. **Methods.** This was a single-centre observational report. Data were abstracted from the electronic health record by two independent reviewers and cross-checked, with discrepancies resolved by consensus. Adverse event terms were coded to MedDRA preferred terms (version 27.0). Causality was assessed with the WHO-UMC system and, independently, with the Naranjo probability scale. Written informed consent was obtained from the patient for publication of anonymised clinical details and images. Laboratory reference ranges are those of the reporting institution's accredited laboratory. Concomitant medicines, over-the-counter products and herbal preparations were recorded where documented. Baseline organ function was reviewed and prior exposure to the suspect class was sought from the dispensing history. The report follows the CARE guideline for case reports. No external funding was received and the authors declare no

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Case 1. An 78-year-old man with atrial fibrillation and stable chronic kidney disease

was started on Cardizen 200 mg twice daily on 5 January 2025. A baseline ECG showed a QTc of 430 ms. On day 6 he had a presyncopal episode; the ECG showed a QTc of 540 ms with frequent ventricular ectopy. Serum potassium was 3.4 mmol/L and magnesium 0.62 mmol/L. Cardizen was stopped and electrolytes were repleted; the QTc returned to 445 ms over 72 hours and there were no further arrhythmias.

Case 2. An 83-year-old woman with heart failure was prescribed Cardizen 150 mg twice daily on 18 February 2025 while also taking a macrolide antibiotic for a chest infection. On day 3 she developed palpitations; telemetry captured a short run of torsades de pointes that terminated spontaneously. The QTc was 560 ms. Cardizen and the macrolide were withdrawn, magnesium sulfate was given, and she was monitored on the coronary care unit. She recovered fully.

Discussion. Methods. This was a single-centre observational report. Data were abstracted from the electronic health record by two independent reviewers and cross-checked, with discrepancies resolved by consensus. Adverse event terms were coded to MedDRA preferred terms (version 27.0). Causality was assessed with the WHO-UMC system and, independently, with the Naranjo probability scale. Written informed consent was obtained from the patient for publication of anonymised clinical details and images. Laboratory reference ranges are those of the reporting institution's accredited laboratory. Concomitant medicines, over-the-counter products and herbal preparations were recorded where documented. Baseline organ function was reviewed and prior exposure to the suspect class was sought from the dispensing history. The report follows the CARE guideline for case reports. No external funding was received and the authors declare no competing interests. Methods. This was a single-centre observational report. Data were abstracted from the electronic health record by two independent reviewers and cross-checked, with discrepancies resolved by consensus.

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1. Roden DM. N Engl J Med. 2004.